

Name: _____ Date: _____

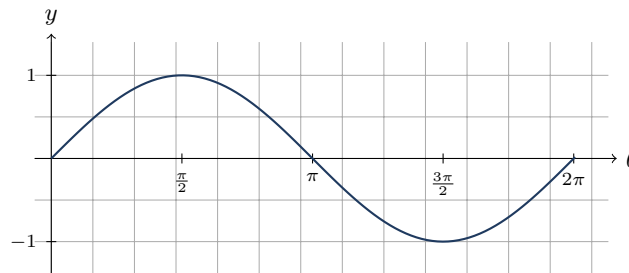
Period: _____

Spiral Review

1. Evaluate each expression using your knowledge of the unit circle.

(a) $\sin\left(\frac{\pi}{2}\right) = \underline{\hspace{2cm}}$ (b) $\cos(\pi) = \underline{\hspace{2cm}}$ (c) $\sin\left(\frac{3\pi}{2}\right) = \underline{\hspace{2cm}}$ (d) $\cos(0) = \underline{\hspace{2cm}}$

2. The graph of $y = \sin \theta$ is shown below with key features labeled. Fill in each blank.



Amplitude = _____ Period = _____ Midline: $y = \underline{\hspace{2cm}}$

3. A function f is transformed by a vertical stretch of factor 4 and a vertical shift up 2. If the parent function has the rule $f(\theta) = \sin \theta$, write the rule of the transformed function.

Transformed function: $f(\theta) = \underline{\hspace{4cm}}$

4. The function $g(\theta) = \sin(\theta - \pi)$ is a horizontal translation of $y = \sin \theta$.

(a) In which direction is the graph shifted? _____

(b) By how many units? _____